Name: Diya Shah

Topic: Active Directory Certificate Services (AD CS) Practicald

1. **Install the Active Directory Certificate Services (AD CS) role on your Windows Server VM and take a screenshot of the 'Role Services' selection screen**.
   - ⇒ Open Server Manager.
     Click Manage → Add Roles and Features.
     Click Next until Server Roles.
     Check Active Directory Certificate Services.
     Click Add Features.
     Click Next.
     On the Role Services page, select:
     Certification Authority
     Click Next.
     Click Install

2. **Configure a new Standalone Root Certification Authority (CA) using the AD CS wizard and set the common name to 'CampusAppCA'**.
   - ⇒ After installation, click the notification flag.
     Select Configure Active Directory Certificate Services.
     Select your server.
     Choose:
     Certification Authority
     Setup Type
     Choose:
     Standalone CA
     CA Type
     Choose:
     Root CA
     Private Key
     Choose:
     Create a new private key
     Cryptography
     Leave the default settings.
     Common Name
     Enter:
     CampusAppCA
     Accept the default validity period.
     Click Configure.

3. **Issue a user certificate for a test user account (e.g., 'testuser') from your CA and export the certificate as a .cer file.**

⇒ Step 1 – Create Test User
Open
Active Directory Users and Computers
Create a user:
Username:
testuser
Step 2 – Open Certification Authority
Go to
Server Manager
↓
Tools
↓
Certification Authority
Step 3 – Certificate Template
Enable the User certificate template if needed.
Step 4 – Request Certificate
Open
MMC
↓
Certificates
↓
Current User
↓
Personal
↓
Certificates
Request a new certificate.
Choose
User
Complete the wizard.
Step 5 – Export Certificate
Right-click certificate
All Tasks
↓
Export
Choose
No Private Key
Save as
testuser.cer

4. **Simulate a Flipkart-style secure login: Configure your server to require client certificate authentication for a test web application and attempt to log in using the issued certificate.

Hint: You can use IIS and map client certificates to user accounts for this demo.**

⇒ Step 1
Install IIS
Server Manager
↓
Add Roles
↓
Web Server (IIS)

⇒ Step 2
Create a simple webpage
Example
C:\inetpub\wwwroot
Create
index.html

⇒ Step 3
Open IIS Manager
Server Manager
↓
Tools
↓
Internet Information Services (IIS)

⇒ Step 4
Select
Default Web Site

⇒ Step 5
Open
SSL Settings
Check
Require SSL

⇒ Step 6
Open
Authentication
Enable

⇒ Client Certificate Mapping Authentication
Step 7
Map
testuser Certificate
↓
testuser Account

⇒ Step 8
Browse

https://ServerName
Browser requests
Select Certificate
Choose
testuser

5. **Use ChatGPT to generate a step-by-step checklist for troubleshooting common AD CS installation errors, and apply one of the steps to intentionally misconfigure your CA, then fix it using the checklist.**
   - ⇒ Step 1
     Verify you are logged in as an Administrator.
     Step 2
     Confirm Active Directory Certificate Services is installed.
     Step 3
     Check required Windows services:
     Remote Procedure Call (RPC)
     Windows Event Log
     DCOM Server Process Launcher
     Step 4
     Open Event Viewer and check:
     Windows Logs → Application
     Windows Logs → System
     Step 5
     Verify the CA configuration:
     CA Type
     Private Key
     Common Name
     Step 6
     Restart the Active Directory Certificate Services service.
     Step 7
     Verify CampusAppCA appears in the Certification Authority console.
     Intentional Misconfiguration
     While configuring AD CS, select:
     Subordinate CA
     instead of
     Standalone Root CA
   - ⇒ Result:
     The configuration fails because a Subordinate CA requires an existing parent CA.
   - ⇒ Fix
     Open Configure Active Directory Certificate Services.
   - ⇒ Select:
     ✓ Standalone CA
     ✓ Root CA

Create a new private key.
Enter the common name:
CampusAppCA
Complete the wizard.
Verify that CampusAppCA appears in the Certification Authority console.
Result:
The CA is successfully configured, and certificate services work correctly.